

Object Oriented and Programming

Lab 4
CLASSES AND OBJECTS (Java)

Objectives

- Creating classes and Objects
- Attributes and members of a class

Objects

- Object: **Objects have states and behaviors.**
Example: A dog has states-color, name, breed as well as behaviors -wagging, barking and eating. **An object is an instance of a class.**
- If we consider the real-world we can find many **objects around us, Cars, Dogs, Humans etc.** All these objects have a state and behavior.

Classes

- **Class**: A class can be defined as a template/ design that describe the **behaviors/states** that object of its type support.

```
public class Dog{  
    String breed;  
    int age;  
    String color;  
  
    void barking() {  
    }  
  
    void hungry() {  
    }  
  
    void sleeping() {  
    }  
}
```

Classes

- A class can contain any of the following variable types.
- Local variables: **variables defined inside methods, constructors or blocks are called local variables.**
The variable will be declared and initialized within the method and the variable will be destroyed when the method has completed.
- Instance variables: **Instance variables are variables within a class but outside any method.** These variables are instantiated when the class is loaded. Instance variables can be accessed from inside any method, constructor or blocks of that particular class.
- Class variables: Class variables are variables declared within a class, outside any method, **with the static keyword.**

Constructor

- Every class has a constructor. If we do not explicitly write a constructor for a class the java compiler builds a default constructor for that class.
- Each time a new object is created at least one constructor will be invoked.
- The main rule of constructors is that they should have the same name as the class.
- A class can have more than one constructor.

Constructor Example

```
public class Puppy{  
    public puppy(){  
    }  
  
    public puppy(String name){  
        // This constructor has one parameter, name.  
    }  
}
```

Creating Objects

- Creating an Object: As mentioned previously a class provides the design for objects. So basically an object is created from a class. In java the new key word is used to create new objects.
- There are three steps when creating an object from a class:
- **Declaration** . A variable declaration with a variable name with an object type.
- **Instantiation** . The '**new**' key word is used to create the object.
- **Initialization** . The 'new' keyword is followed by a call to a constructor. This call initializes the new object.


```
public class MyClass {  
    int x;  
  
    public MyClass(int y) {  
        x = y;  
    }  
  
    public static void main(String[] args) {  
        MyClass myObj = new MyClass(5);  
        System.out.println(myObj.x);  
    }  
}
```

Creating Objects

```
public class Puppy{  
  
    public Puppy(String name){  
        // This constructor has one parameter, name.  
        System.out.println("Passed Name is :" + name );  
    }  
  
    public static void main(String []args){  
        // Following statement would create an object myPuppy  
        Puppy myPuppy = new Puppy( "tommy" );  
    }  
}
```

```
public class PAge {
    int petAge;

    public PAge(String name)
    {
        System.out.println("Passed Name is"+ name);
    }

    public void setAge (int age)
    {
        petAge = age;
    }

    public int getAge()
    {
        System.out.println("Puppy's age is:"+petAge);
        return petAge;
    }

    public static void main (String [] args)
    {
        PAge myPet = new PAge("Tommy");

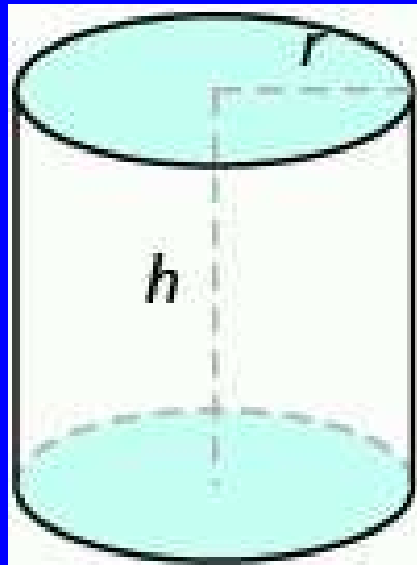
        myPet.setAge(2);

        myPet.getAge();

        System.out.println("Variable Value:"+myPet.petAge);
    }
}
```

Lab Tasks

1. Compile and execute the code at point 9 in netbeans and show the output.
2. Create a code using classes to calculate the surface area and volume of a cylinder. Show code and output here. You need to get the radius input from the user



Volume: $V = \pi r^2 h$ or $V = Bh$

Surface Area: $S = 2\pi r^2 + 2\pi rh$

```
public class MainCls {
```

```
    static double r=4;
```

```
    static double h=6;
```

```
    public static void main (String args[])
```

```
    {
```

```
        Cylinder obj1 = new Cylinder();
```

```
        obj1.area(r, h);
```

```
        obj1.volume(r, h);
```

```
    }
```

```
}
```

```
package joptionpan;
```

```
/**
```

```
 *
```

```
 * @author Administrator
```

```
 */
```

```
public class Cylinder {
```

```
    double pi=3.14;
```

```
    public void volume (double r, double h)
```

```
    {
```

```
        System.out.println("The Vaolum is:"+pi*r*r*h);
```

```
    }
```

```
    public void area (double r, double h)
```

```
    {
```

```
        System.out.println("The Area is: "+(2*(pi*r*r)+2*pi*r*h));
```

```
    }
```

```
}
```

Example 2 - Class Car

- class Car {
- // Attributes (fields)
- String brand;
- String color;
- int year;
-
- void displayInfo() {
- System.out.println("Brand: " + brand);
- System.out.println("Color: " + color);
- System.out.println("Year: " + year);
- }
- }

- **public class Main {**
- **public static void main(String[] args) {**
- **Car car1 = new Car();**
- car1.brand = "Toyota";
- car1.color = "Red";
- car1.year = 2020;
-
- Car car2 = new Car();
- car2.brand = "Honda";
- car2.color = "Blue";
- car2.year = 2022;
-
- System.out.println("Car 1 Details:");
- car1.displayInfo();
-
- System.out.println("\nCar 2 Details:");
- car2.displayInfo();
- }
- }